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VINTAGE SIGNAL PROCESSORS · REISSUED IN LOGIC

DATA SHEET

G2 PROGRAMS
Revision 1.0

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G2 Amiga release 1.0 · 2026-11-03

AmigaOS Programs of the G2 Amiga

The monitors, drivers and helpers that the release installs on the Amiga: serial and MIDI monitors, the FT2232H and configuration-flash tools, bsdsocket.library, the USBStick: volume, the serial and flash devices and the AHI driver.

HIGHLIGHTS

- Four monitor programs in one house style.
- bsdsocket.library on the firmware: FTP and SSH clients run.
- One floppy image with an Install script.

KEY PARAMETERS

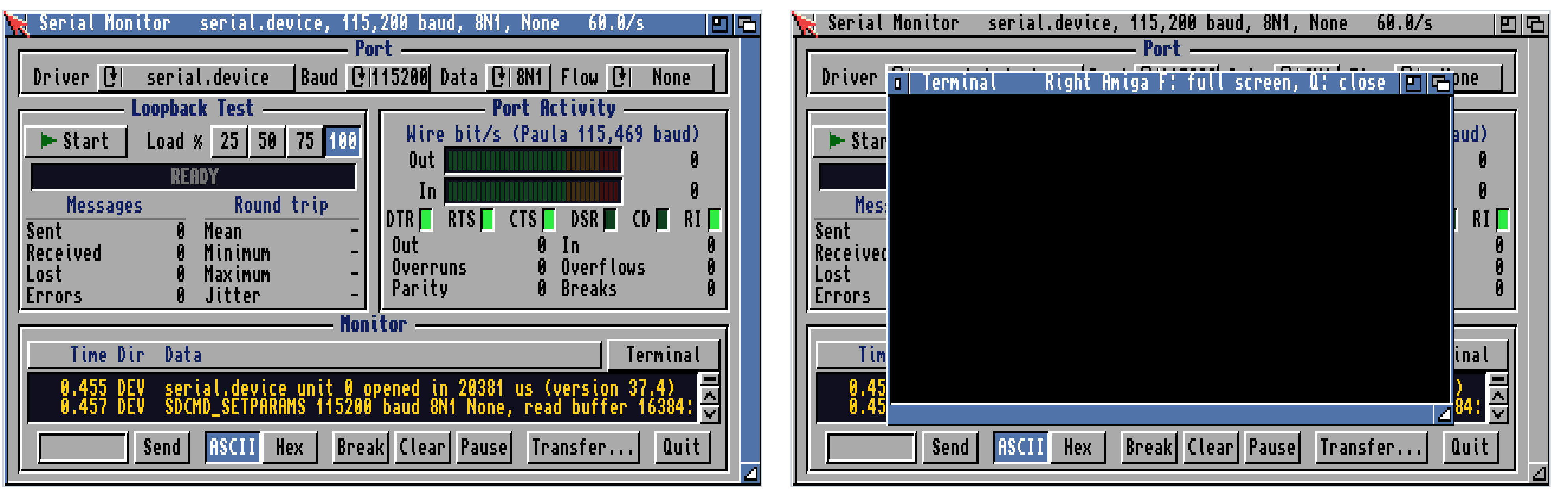
PARAMETER	VALUE
Programs	7 in C:
Drivers	4 in DEVS:
Handler	L:share-handler
CPU	68020
Smallest screen	640 × 200

1 PROGRAMS AND DRIVERS

FILE	INSTALL	BYTES	FUNCTION	LICENCE
serialmon	C:	164,028	Serial port monitor: loopback test, meters, modem lines, monitor, driver switching, ZMODEM/YMODEM/XMODEM, VT100/VT220 terminal	PolyForm NC; libvterm MIT
midimon	C:	66,964	MIDI monitor, loopback tester and message sender on Paula's port	PolyForm NC
ftdimon	C:	207,532	FT2232H: device, pins, JTAG chain, SPI NOR flash, hex editor	GPL-2.0-or-later
flashmon	C:	76,128	The G2 configuration flash: part, map, read, verify, write, erase, CRC, bitstream extraction, hex editor	GPL-2.0-or-later
PanoTCP	C:	34,808	bsdsocket.library 4 on the firmware's TCP/IP stack	PolyForm NC
PANOFS	C:	856	Inhibits DH0: while the firmware writes into its image	PolyForm NC
lsz, lrz	C:	94,248, 97,532	ZMODEM, YMODEM, XMODEM send and receive (lrzsz 0.12.20)	GPL-2.0
share-handler	L:	13,560	The USBStick: volume (with DEVS:DOSDrivers/SHARE); from a314fs	CC0-1.0 origin, PolyForm NC
g2serial.device	DEVS:	11,992	serial.device command set on the serial FIFO board or Paula	PolyForm NC
panoflash.device	DEVS:	4,108	The configuration flash through HD0's flash window	GPL-2.0-or-later
g2.audio, G2	DEVS:AHI, DEVS:AudioModes	4,952, 408	AHI driver and mode file for the Toccata path	AHI driver terms, PolyForm NC

All are on G2TOOLS.adf with an Install script. Built with m68k-amigaos-gcc 6.5.0b for the 68020; every window follows the screen of the Workbench and fits 640 × 200.

2 SERIALMON


Serial Monitor   serial.device, 115,200 baud, 8N1, None   60.0/s

Driver serial.device   Baud 115200   Data 8N1   Flow None

READY

Wire bit/s (Paula 115,469 baud)

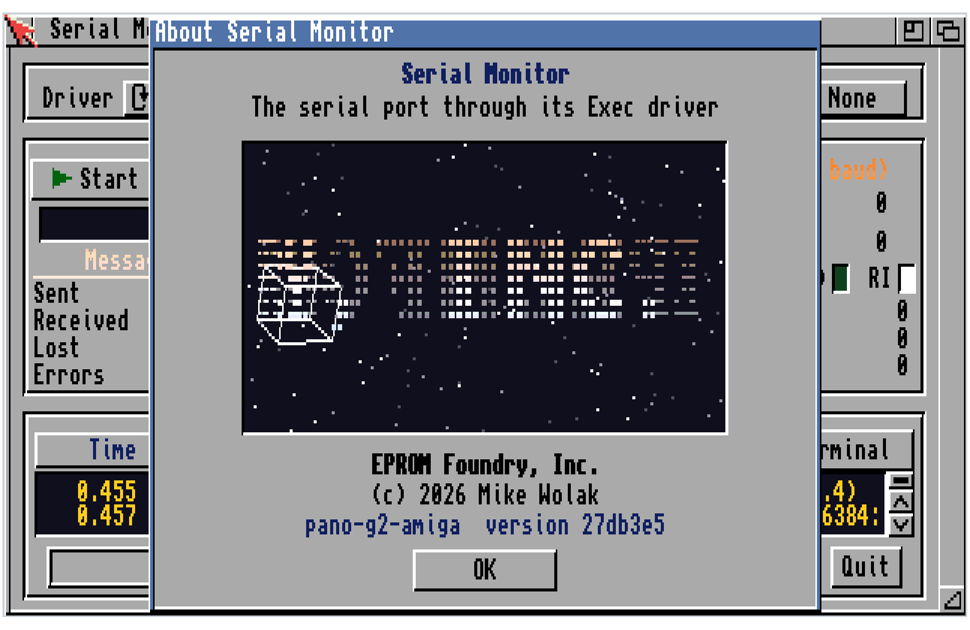
```
Sent         0   Mean      -     Out        0   In          0
Received     0   Minimum   -     Overruns   0   Overflows   0
Lost         0   Maximum   -     Parity     0   Breaks      0
Errors       0   Jitter    -
```

```
0.455 DEV  serial.device unit 0 opened in 20381 us (version 37.4)
0.457 DEV  SDCMD_SETPARAMS 115200 baud 8N1 None, read buffer 16384:
```

Terminal   Right Amiga F: full screen, Q: close

Port
Loopback Test
Start
Load %
25
50
75
100
Messages
Round trip
Port Activity
Out
In
DTR
RTS
CTS
DSR
CD
RI
Monitor
Time Dir Data
Terminal
Send
ASCII
Hex
Break
Clear
Pause
Transfer...
Quit

Figure 1. serialmon: the main window (driver, rates, loopback test, port activity, monitor) and the terminal window (Right Amiga V; Right Amiga F full screen).


About Serial Monitor

Serial Monitor
The serial port through its Exec driver

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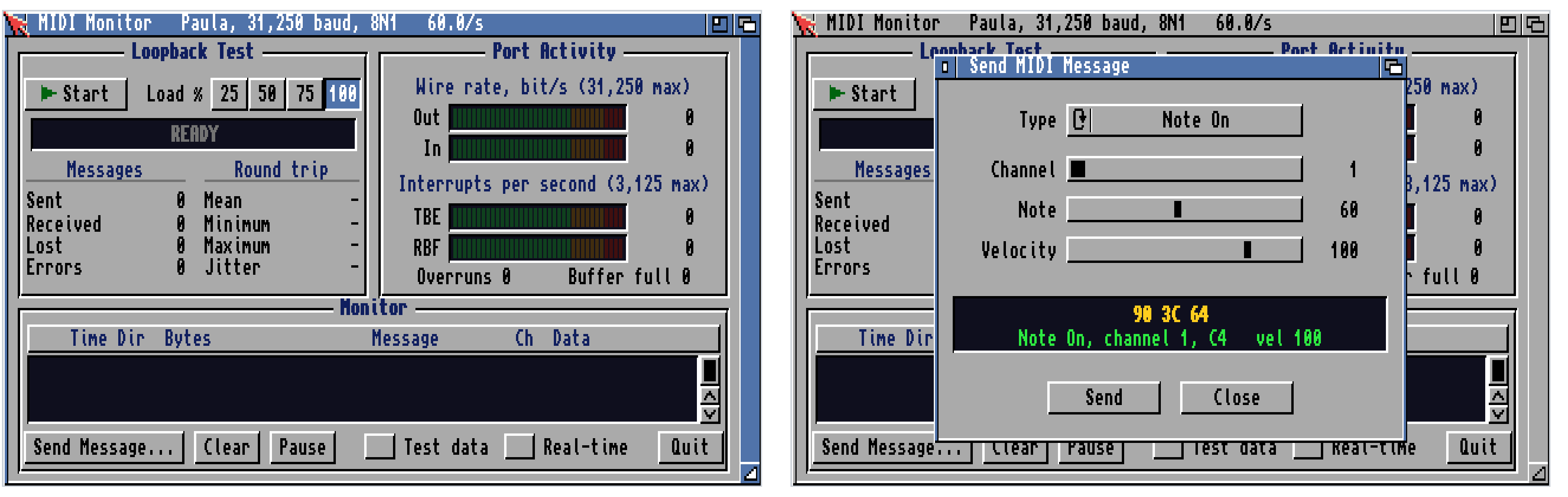
pano-g2-amiga version 27db3e5

Figure 2. The About box: the logo on a turning sphere with a cube and a starfield, played by DMA.

3 MIDIMON


MIDI Monitor   Paula, 31,250 baud, 8N1   60.0/s

READY

Wire rate, bit/s (31,250 max)

Interrupts per second (3,125 max)

```
Sent         0   Mean      -     TBE   0
Received     0   Minimum   -     RBF   0
Lost         0   Maximum   -     Overruns 0     Buffer full 0
Errors       0   Jitter    -
```

Send MIDI Message

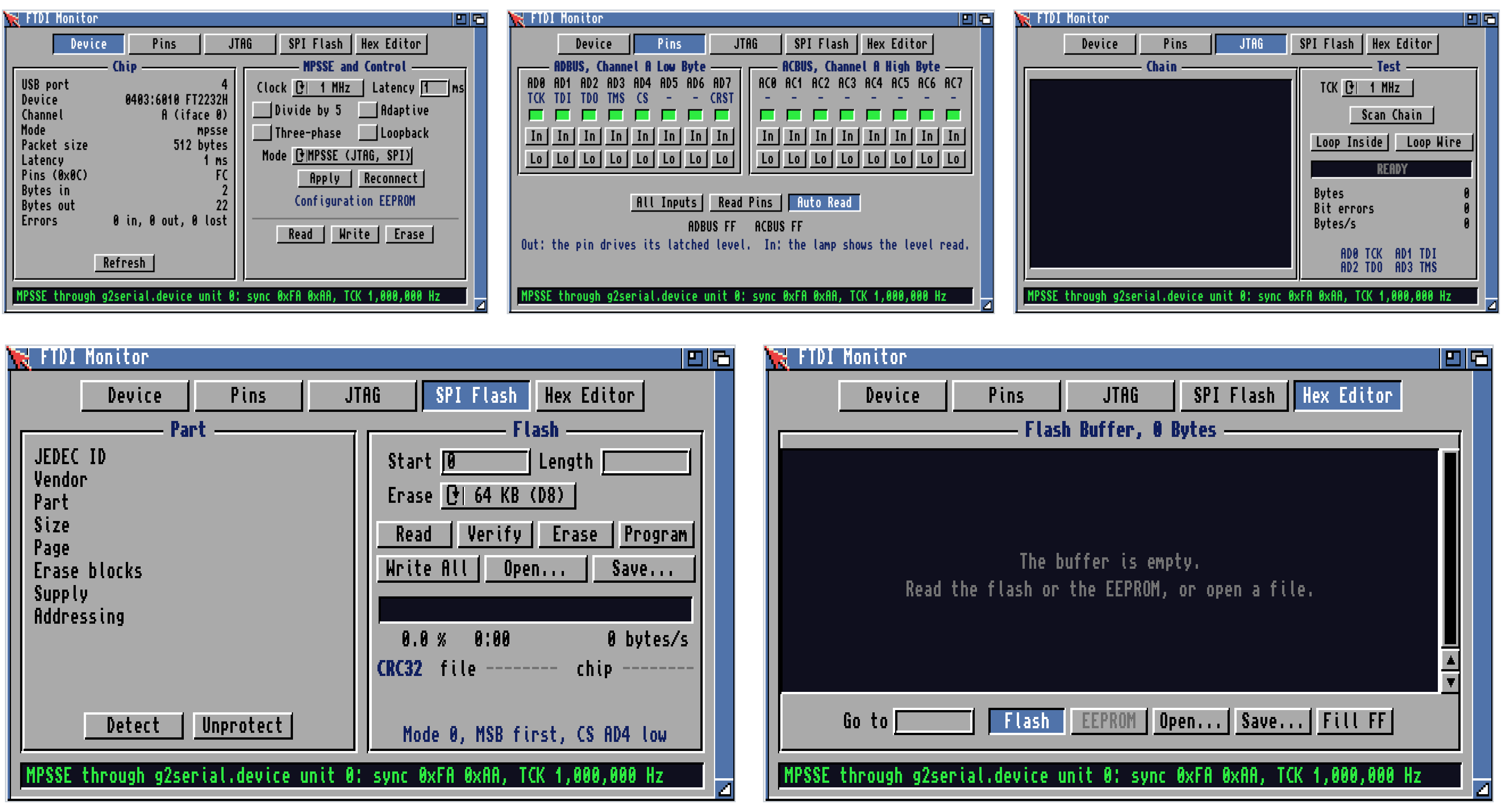
Type Note On
Channel 1
Note 60
Velocity 100

```
90 3C 64
Note On, channel 1, C4   vel 100
```

Loopback Test
Start
Load %
25
50
75
100
Messages
Round trip
Port Activity
Out
In
Monitor
Time Dir Bytes
Message
Ch Data
Send Message...
Clear
Pause
Test data
Real-time
Quit
Send
Close

Figure 3. midimon: loopback test and port meters at 31,250 baud; the send panel shows only the controls of the chosen message type.

4 FTDIMON


FTDI Monitor

```
USB port                    4
Device       0403:6010 FT2232H
Channel            A (iface 0)
Mode                     mpsse
Packet size          512 bytes
Latency                   1 ms
Pins (0x0C)                 FC
Bytes in                     2
Bytes out                   22
Errors      0 in, 0 out, 0 lost
```

Clock 1 MHz   Latency 1 ms   Mode MPSSE (JTAG, SPI)

```
AD0 AD1 AD2 AD3 AD4 AD5 AD6 AD7     AC0 AC1 AC2 AC3 AC4 AC5 AC6 AC7
TCK TDI TDO TMS CS  -   -   CRST    -   -   -   -   -   -   -   -
```

ADBUS FF   ACBUS FF

Out: the pin drives its latched level. In: the lamp shows the level read.

TCK 1 MHz

READY

```
Bytes        0
Bit errors   0
Bytes/s      0
```

AD0 TCK AD1 TDI
AD2 TDO AD3 TMS

Start 0   Erase 64 KB (D8)

```
0.0 %   0:00        0 bytes/s
CRC32  file --------   chip --------
```

Mode 0, MSB first, CS AD4 low

Flash Buffer, 0 Bytes

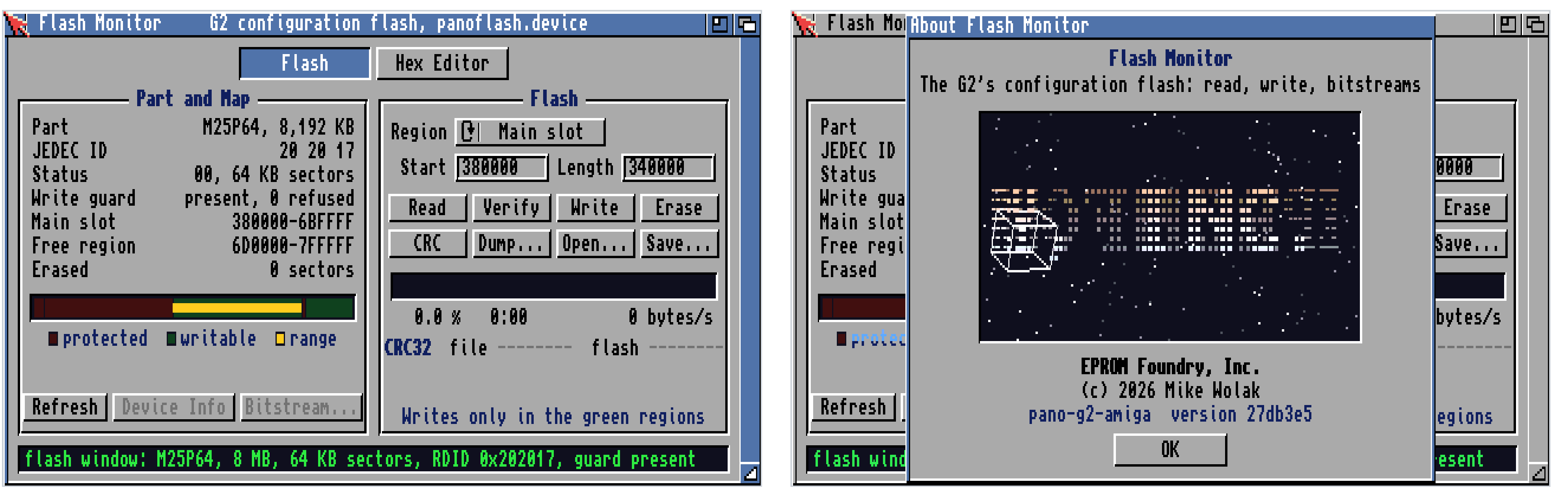
The buffer is empty.
Read the flash or the EEPROM, or open a file.

```
MPSSE through g2serial.device unit 0: sync 0xFA 0xAA, TCK 1,000,000 Hz
```

Device
Pins
JTAG
SPI Flash
Hex Editor
Chip
MPSSE and Control
Divide by 5
Adaptive
Three-phase
Loopback
Apply
Reconnect
Configuration EEPROM
Read
Write
Erase
Refresh
ADBUS, Channel A Low Byte
ACBUS, Channel A High Byte
In
Lo
All Inputs
Read Pins
Auto Read
Chain
Test
Scan Chain
Loop Inside
Loop Wire
Part
JEDEC ID
Vendor
Part
Size
Page
Erase blocks
Supply
Addressing
Detect
Unprotect
Flash
Length
Verify
Program
Write All
Open...
Save...
Go to
Flash
EEPROM
Fill FF

Figure 4. ftdimon pages Device, Pins, JTAG, SPI Flash and Hex Editor, an FT2232H on hub port 4 in MPSSE mode.

5 FLASHMON


Flash Monitor   G2 configuration flash, panoflash.device

```
Part            M25P64, 8,192 KB
JEDEC ID                20 20 17
Status          00, 64 KB sectors
Write guard     present, 0 refused
Main slot           380000-6BFFFF
Free region         6D0000-7FFFFF
Erased                  0 sectors
```

Region Main slot   Start 380000   Length 340000

```
0.0 %   0:00        0 bytes/s
CRC32  file --------   flash --------
```

Writes only in the green regions

```
flash window: M25P64, 8 MB, 64 KB sectors, RDID 0x202017, guard present
```

About Flash Monitor

Flash Monitor
The G2's configuration flash: read, write, bitstreams

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Flash
Hex Editor
Part and Map
Flash
protected
writable
range
Refresh
Device Info
Bitstream...
Read
Verify
Write
Erase
CRC
Dump...
Open...
Save...
OK

Figure 5. flashmon: the M25P64 with its map (protected regions red, writable green) and the About box.

6 CAPTURES

Every figure is a frame of the unit's video capture, taken with the firmware console command `snap` (640 RGB565 samples a line) on an NTSC 640 × 200 Workbench screen, each line shown twice.

Captured on a G2 revision C unit running release 1.0 (firmware 0x27DB3E56, fabric 0x0AB5EF2D). Repository mikewolak/pano-g2-amiga, release 1.0.